



DRIVING CHANGE:

KEY INSIGHTS FROM CAR

ACCIDENT TRENDS IN LOS ANGELES

Meet the Team

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Statistical Analysis & Interpretation



All team members contributed to data validation, insight review, and final conclusions.

Analysis Objectives

- Reduce Traffic Accidents
- Improve Public Safety
- Improve Data-Driven Urban Planning



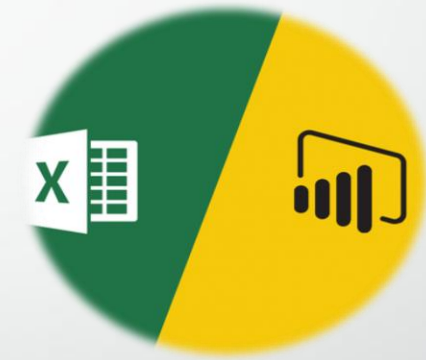
Dataset Overview



68,956 traffic accident records
March 2016 – December 2021



Data source: Los Angeles Police
Department (LAPD)



Data preparation:
Raw XLSX files cleaned, transformed,
and modeled using **Excel** and **Power BI**)

Data was validated for missing values, inconsistent timestamps, and categorical anomalies.

Key Analytical Questions

To support traffic safety improvements in Los Angeles, we focused on the following key questions:

- At what time of day do most traffic accidents occur?
- When do the most severe accidents happen (daylight vs night, weekday vs weekend)?
- Do weather conditions influence accident frequency or severity?



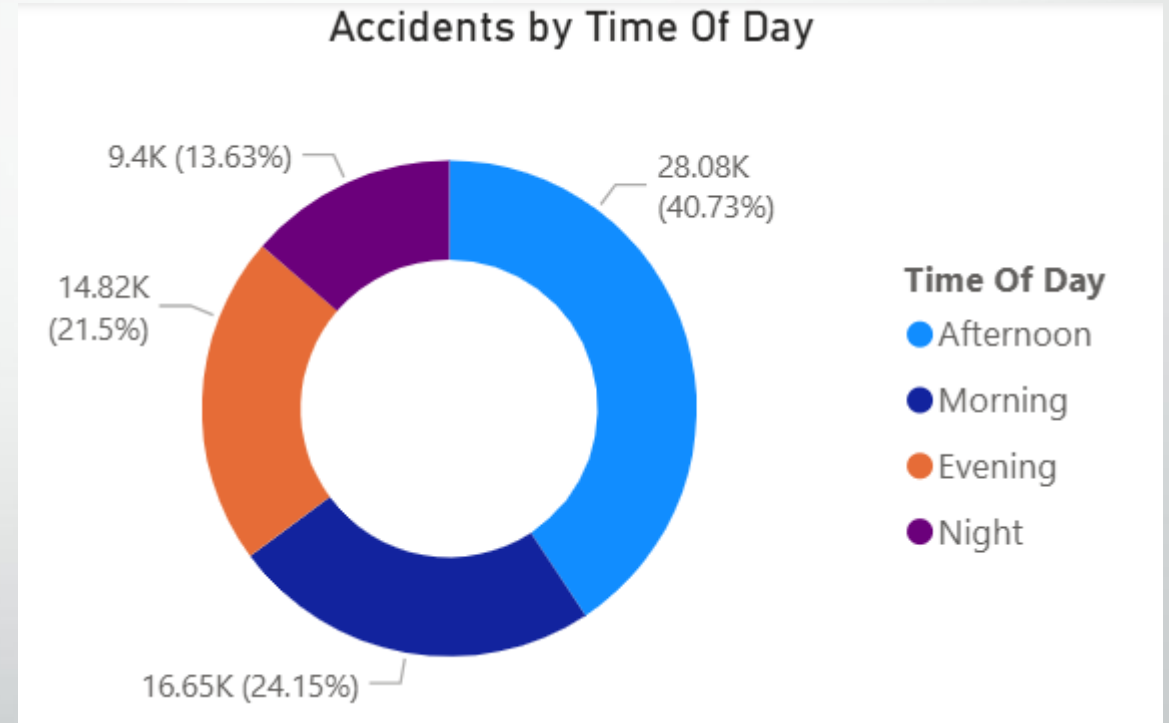
Traffic Accident Volume Over Time (2016–2021)



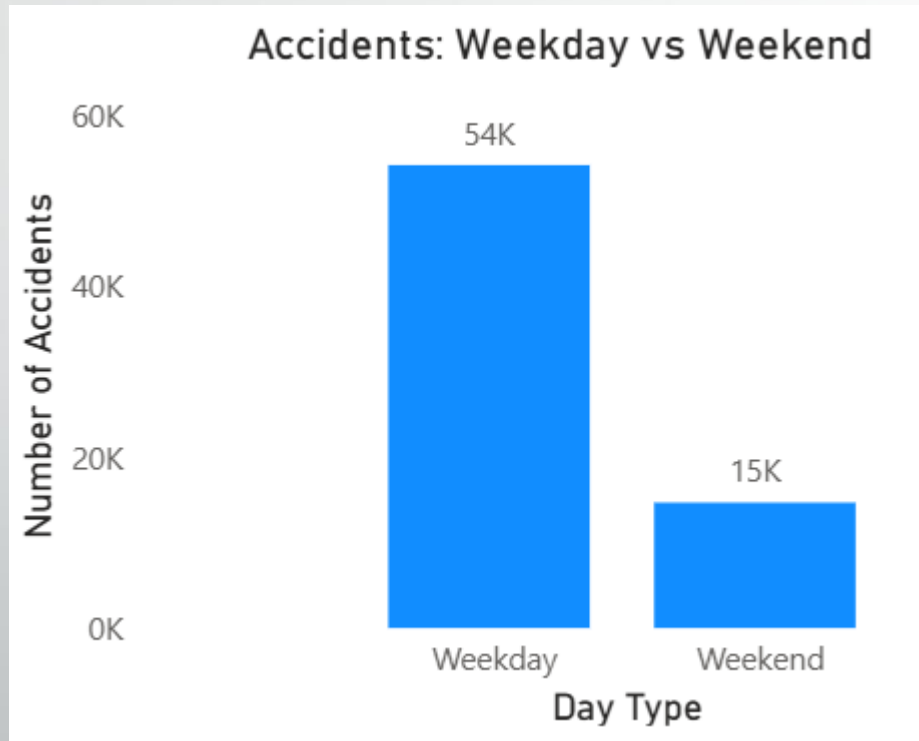
- Traffic accidents remained relatively stable between **2016 –2018**
- A **significant upward trend** is observed from **2019 onward**
- This trend highlights increasing pressure on urban traffic systems

Accidents by Time of Day

- **Afternoon** accounts for the largest proportion of accidents (~41%)
- **Morning and evening** show similar but lower accident volumes



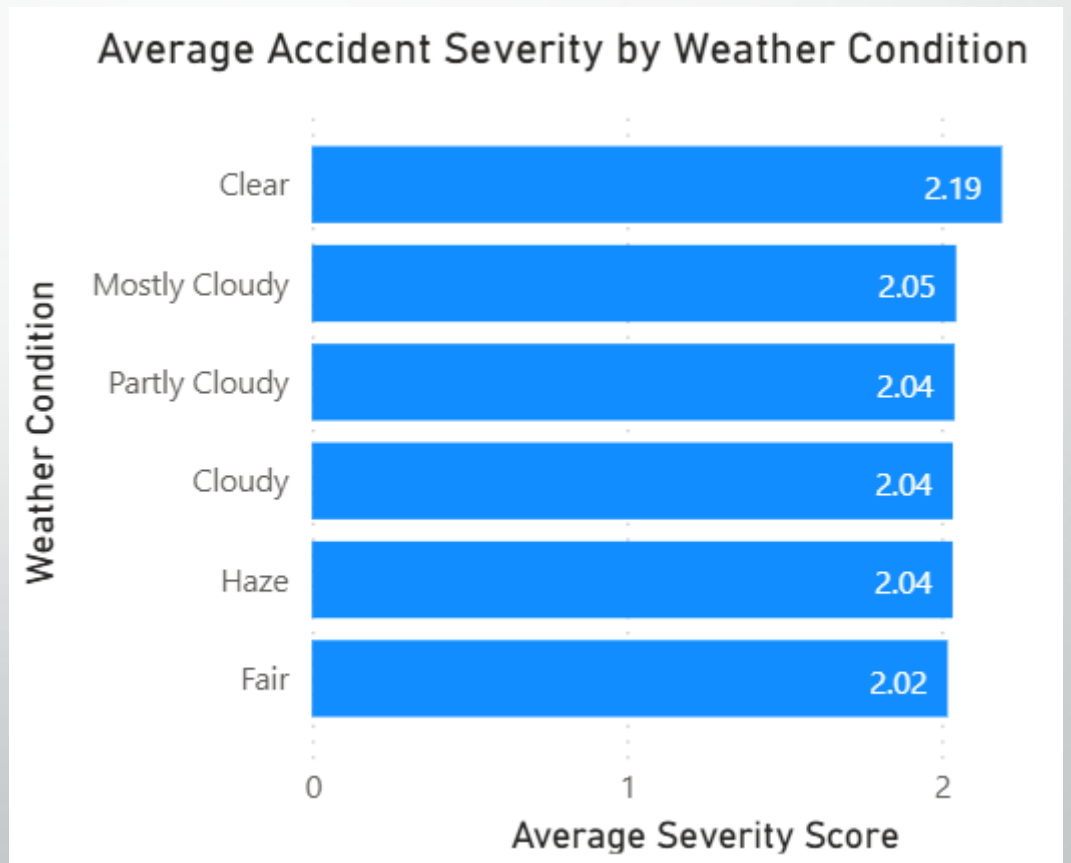
Accident Severity by Time and Day Type

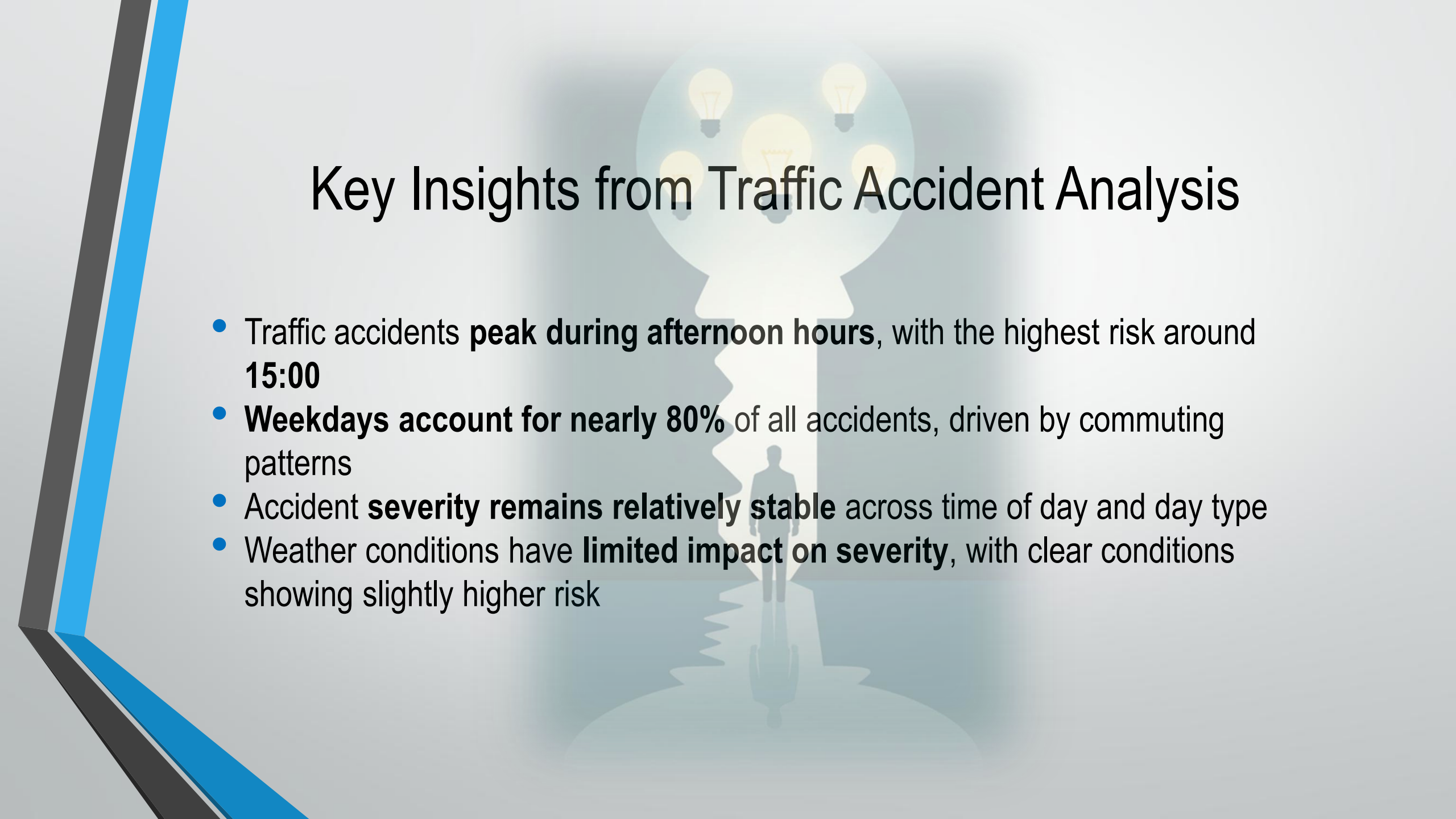


- Average accident severity is **similar on weekdays and weekends**
- No extreme severity spikes are observed across time periods

Impact of Weather Conditions on Accident Severity

- **Clear and overcast conditions** show slightly higher average severity
- **Rain and fog** do not significantly increase severity levels



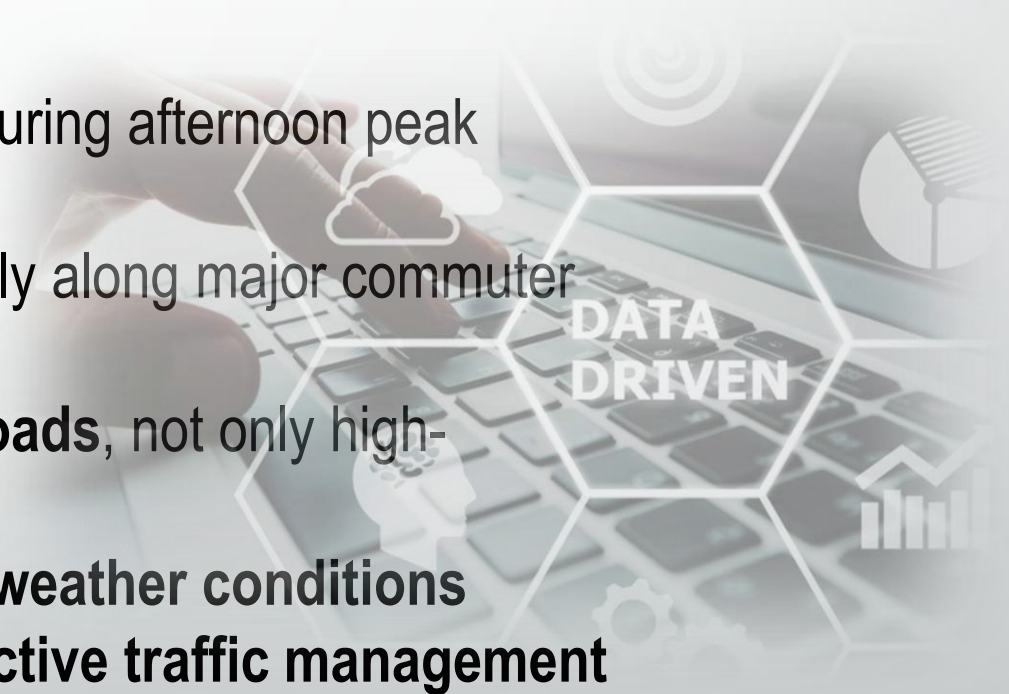


Key Insights from Traffic Accident Analysis

- Traffic accidents **peak during afternoon hours**, with the highest risk around **15:00**
- **Weekdays account for nearly 80%** of all accidents, driven by commuting patterns
- Accident **severity remains relatively stable** across time of day and day type
- Weather conditions have **limited impact on severity**, with clear conditions showing slightly higher risk

Data-Driven Recommendations

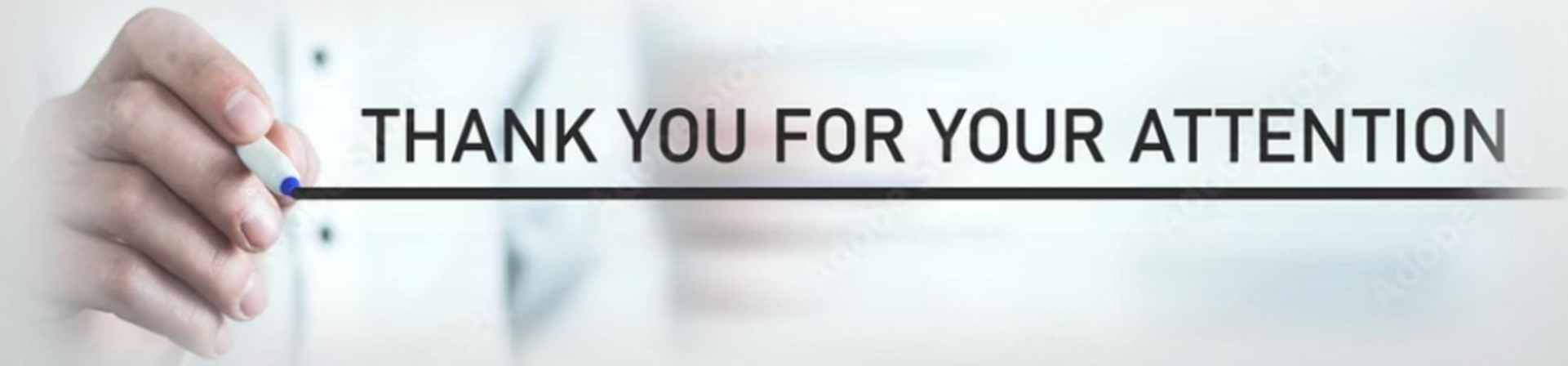
- **Target traffic monitoring and enforcement** during afternoon peak hours (15:00–18:00)
- **Optimize traffic flow on weekdays**, particularly along major commuter corridors
- Prioritize safety interventions on **high-traffic roads**, not only high-severity locations
- Maintain traffic safety measures even in **clear weather conditions**
- Use historical accident trends to support **predictive traffic management**



Conclusion

- Traffic accident risk in Los Angeles is **primarily driven by traffic volume and human behavior**
- The **afternoon period on weekdays** represents the highest overall risk
- Accident severity shows **limited variation** across time, day type, and weather conditions
- Data-driven targeting of **peak hours and high-traffic corridors** offers the greatest potential impact

Thank You



THANK YOU FOR YOUR ATTENTION

Los Angeles Traffic Accident Analysis
Big Blue Academy Datathon | 2025